



ICT

Primary 6

First semester 2026-2027

Name:.....

School:.....

Grade:.....

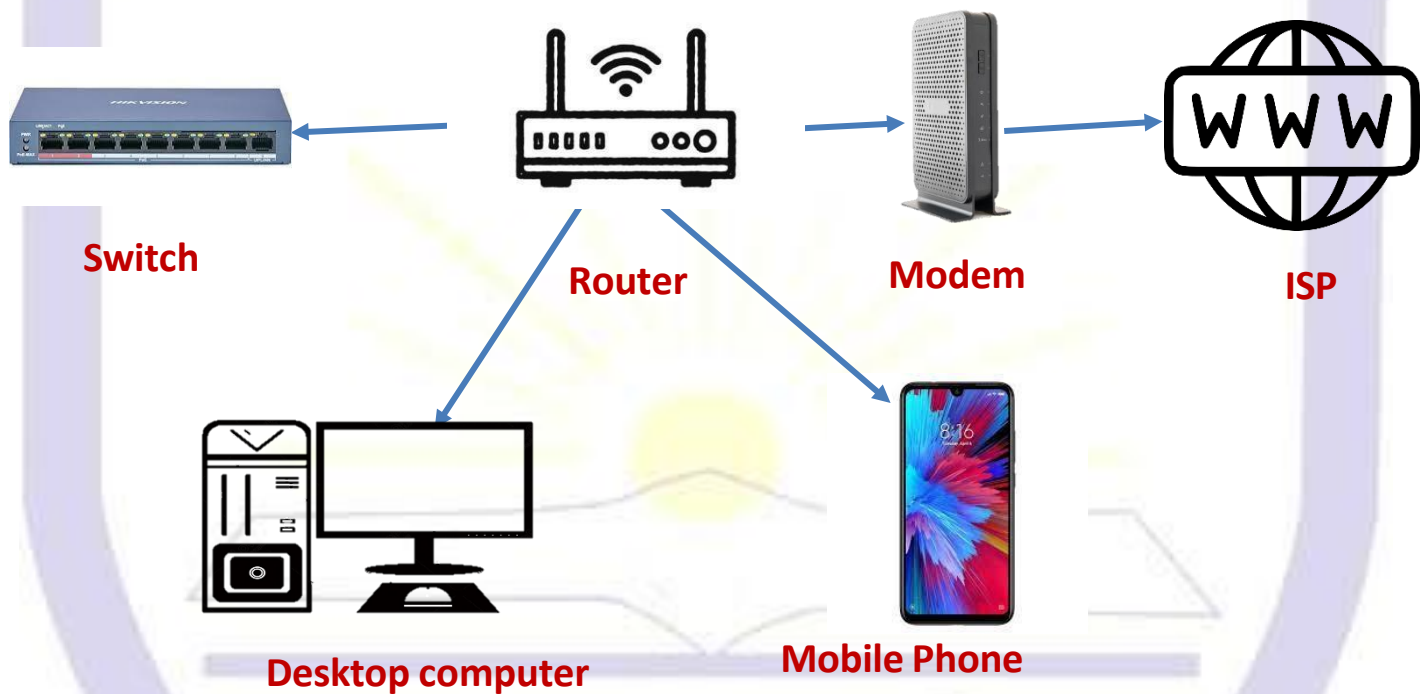
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Computer Network Devices

Network

They are groups of people, or things, which are connected to each other for a common purpose.

How connect devices?



Making a connection

(Wired networks)

Use **physical wiring** and **cables** which are plugged into **holes** on **devices**, called **ports**.



(Wireless networks)

Connected **without wiring** by using **radio waves** or **Wi-Fi**



Computer Network



1

Modem

- A device that connects a local area network (LAN) to internet.
- It converts and transforms the signal from the internet service provider (ISP) into a digital signal can be read by devices



2

Switches

- Devices can send data to a specific device over a network.
- It helps to make connections easier and faster.



3

Router

- A device that distributes the Internet everywhere in the house.
- Whether wired or wireless (WI-FI)



4

ports

- Some devices connect to the Internet through special wires. These wires are inserted into holes called **ports**.

5 LAN network



A local network connected by a cable between computers is a type of wired network.

Artificial Intelligence



They are computer applications that perform complex tasks that simulate the thinking and performance of the human element.

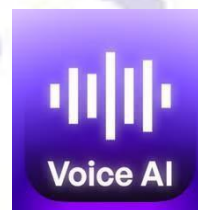
Uses of Artificial Intelligence

The ability to learn and reason, by providing these technologies with information and knowledge, and possible answers to all possible questions, the machine can give answers to the data provided, such as:

- **Writing an article on a topic** , Inquiring about historical events and their sequence.



Chat GPT



Axis 1

Exercise

1

1- Put (✓) or (X):

- 1- Connection between networks is established by wired cables only. ()
- 2- Computers need different devices to connect to the network. ()
- 3- Artificial intelligence cannot simulate human thinking. ()
- 4- Cell phones are used to connect the computer network to the Internet Service Provider (ISP). ()
- 5- Wired Networks are connected via Wi-Fi. ()
- 6- It is impossible to connect a tablet to the Internet. ()

2- Choose the correct answer:

- 1- can be wirelessly connected to the network.
(a) Typewriter (b) Calculator (c) Tablet
- 2- The holes in devices where wired network wires and cables are connected are called
(a) Radio waves (b) Wi-Fi (c) Ports
- 3-are considered intelligence components of a computer network.
(a) Switches (b) Printers (c) Desktops
- 4- Computer applications that perform tasks that mimic human thinking and performance are called.....
(a) Cybersecurity (b) artificial intelligence (c) computer networks
- 5- A group of devices or things connected together is called.....
(a) Networks (b) Computer (c) Keyboard (d) Mobile devices
- 6- sends data to a specific device over a network.
(a) Computer (b) Artificial intelligence (c) Switch (d) Tablet

Augmented reality, virtual reality, and artificial intelligence feel futuristic, but they are already being used in education and everyday life.

Virtual reality (VR)

It is a simulated 3-D environment that allows users to explore and interact with the surroundings, as if it were reality, users often wear a VR headset.

Examples

In the classroom

VR can take you to a world where bigger than you, so that you can

In historical sites

It can take you to museum thousands of miles away, or to a historical site hundreds of years ago study them closely.



Augmented reality (AR)

- It combines the real world with a virtual world by adding computer-generated images, by the camera of a smartphone or a tablet and will show a 3-D model of what looks like the real world.

- AR helps students become active participants in learning and makes learning more memorable.

Examples

In Math class

Students in a math class can point their smartphone camera at a flat surface and project a 3-D image like a cube onto surface study them closely.



Artificial intelligence (AI)



- It is the ability of a machine such as a computer to think and learn, a computer learns by analyzing provided examples.
AI is used to unlock your phone with facial recognition and to personalize social media content and search results



How can futuristic technology improve education and everyday life

- Someone with a visual impairment can use **AR** to see their family members up close.
- Disabled individuals can use **AI** to accomplish everyday tasks (making phone calls and navigating their computer using their voice or sign language).
- **VR** can immerse able-bodied people in experiences that demonstrate the barriers disabled people face every day

Axis 1

Exercise

2

Put (✓) or (X):

- 1- Futuristic technology can improve everyday life. ()
- 2- Augmented reality combines the real world with the virtual world. ()
- 3- It is difficult for AI to unlock phones by face. ()
- 4- is a real environment to explore things. ()
- 5- Virtual reality is a type of future technology. ()
- 6- Artificial intelligence is the ability of a computer to think or learn. ()

2- Complete the following sentences using words between brackets:

(Virtual Reality - improving - Augmented Reality – historical Sites - Artificial Intelligence)

- 1- The ability of a computer to think and learn is called.....
- 2- Users wear..... headsets to see another world.
- 3- Futuristic technology..... can improve the lives of people with disabilities.
- 4-..... helps students to be active in the learning process.
- 5- Virtual reality can be used to visit.....

Assistive technology



It helps people of determination with tasks they find challenging.

For example

- Screen magnification software
- Hearing aids
- Programs that turn speech to text and text to speech.
- **Smart gloves** that empower deaf people to translate their signs in real time to spoken or text output
- Braille keyboard helps blind people read and write easily.



It is important to be objective and evaluate these new technologies

User Experience

- User experiences describes how people use and interact with a product, and how they evaluate their experiences.
- User experience is often broken into different areas, and can be positive or negative.



How to evaluate the product?

- 1- **Accessibility**: Is the technology accessible to people with disabilities?
- 2- **Credible**: Are the company and its products trustworthy?
- 3- **Desirable**: Does the product meet your needs?
- 4- **Findable**: If there is an issue with the product or service, can we find a solution?
- 5- **Usable**: Is it easy to use?
- 6- **Useful**: Is the technology practical?



- 7- **Valuable**: Does this technology improve my life or help me learn?



Axis 1

Exercise

3

1- Put (✓) or (X):

- 1- It is important to evaluate a new technology to make the decision to use it. ()
- 2- User experience is not related to the development of products and services. ()
- 3- Advanced assistive technology varies with the goal of improving someone's

independence. ()

4- User experience is divided into different areas and can only be positive. ()

5- Assistive technology helps to integrate people into society. ()

2- Choose the correct answer:

1- The latest and most advanced version of a service or product is called..... technology

(a) assistive (b) cutting-edge (c) late (d) complex

2- Enabling deaf people to translate their gestures into a texture or spoken output is.....

(a) secret technology (b) virtual reality (c) smart gloves (d) microphone

3) Society makes decisions about the use and development of technology through a process of.....

(a) searching (b) information gathering (c) evaluation (d) modernization

4-..... technology that enables people with disabilities to perform tasks they find difficult

(a) Updated (b) Assistive (c) Advanced (d) Inappropriate

5- is considered is an example of cutting-edge technology.

(a) website creation (b) artificial intelligence (c) librarianship (d) other



Axes 1

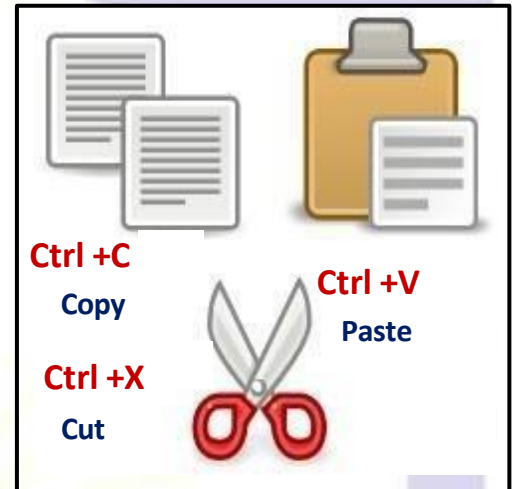
DIGITAL RESEARCH SKILLS

4

Using digital editing tools



The most commonly used word processing commands and are very useful when copying, pasting, or cutting text, information for citing sources, or images into a word processing file.



Digital search tools



Internet searches, specialized search engines and verified online libraries used search for information, but the results will be different



(Search Engines)

- It help people to search internet websites based on the key words and phrases.
- They are user-friendly, very fast, and often result in many results.



(Database)

- They are collections of information which are typically stored in a computer system
- Accessed for free through a school or library.
- good place to search for information from journals, newspapers, and reference books.
- A general database such as The EKB (covers many subjects, and results are credible and accurate).
- A subject database (covers only one subject, such as business, and any results are specific to this topic).



(Library Catalogs)

- It is a database which includes all the items owned by a library. Such as recently published books, e-books
<https://www.bibalex.org>



N.L.S

(Conducting a database search)

- Use a keyword search.
- Review the results for subject headings.
- Use other database fields (author or journal title), to narrow a search.
- Limiting a search to the last few years will help you find the most up-to-date research.

KEYWORDS

Axis 1

Exercise

4

1-Put (✓) or (X):

- 1- Keyboard shortcuts are useful when typing in a text file. ()
- 2- Specialized search engines are a tool for digital research. ()
- 3- You need to determine what results you need to determine which search tool you can use. ()
- 4- The Bibliotheca Alexandrina catalog is an example of a search engine. ()
- 5- The Digital Library Catalog is a database that includes all the library's resources and items. ()
- 6- The transcription tool is one of the tools of digital research. ()
- 7- Different fields can be used in the database such as the author's name ()
- 8- Reliable electronic libraries help in online research ()
- 9- Databases are considered one of the tools of digital editing ()

2- Choose the correct answer:

- 1-a collection of information stored in a computer system
(a) search engines (b) keystrokes (c) databases (d) keyboards
- 2- An example of a database is
(a) Egyptian Knowledge Bank (b) Email (c) Posters (d) Videos
- 3- enables you to perform digital searches within specialized bodies.
(a) printed books (b) paper articles (c) library catalogs (d) interviews
- 4- When searching any database, use in the search.
(a) images (b) numbers (c) keywords (d) wrong words
- 5- The Egyptian Knowledge Bank is an example of a public database
(a) reliable (b) available (c) unreliable (d) inaccurate



Operating System



- Is the software that manages a computer's functions (memory, drives, printers, and other devices).
- It includes accessible devices (screen readers and magnifiers).
- It runs a computer's software and hardware and acts as a translator, allowing a user to communicate with the computer without knowing computer languages.
- Most computers come pre-loaded with an operating system.



Mobile operating systems



Android

- The most popular (72% of the world market)
- uses **open-source** software (means the programming code can be read or changed by anyone).

iOS



- Apple iOS(27%) (which is used on the iPhone).
- uses **closed -source** software (it can't be seen or changed by the public).



Both **Android** and **iOS** perform the same basic functions (messaging, web access video chat and touch function).



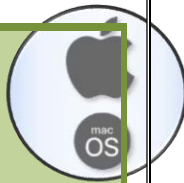
Computer operating systems

Microsoft Windows



- Works on the majority of computers globally (closed – source)

Mac OS



- Works on Apple computers only.



Cutting-edge operating systems

Robot Operating System (ROS):

It is an open-source program for building robotic applications.

The operating systems that power self-driving cars:

These include those developed by Apple Car Play and Windows™ Automotive, as well as private car companies.



Programming languages



- Instructions to a computer are given in code.
- They have evolved from simple machine language (binary) to high-level human-like languages.
- Programmers choose a language based on purpose.
- They are usually open source (Anyone can see, modify, or distribute the code).

- Programmers use Hyper Text Markup Language (HTML) to create web pages.



Computer languages often share similarities, so once you learn one, it is easier to learn others.

HTML

- HTML is commonly chosen for website creation because it is easy to learn and to use.
- It allows creators to add elements (text, links, images, audio, and even videos to their web pages).
- It is easy to make changes and it is compatible with all the major web browsers.





1-Put (✓) or (X):

- 1-The operating system is unable to control auxiliary devices, such as screen readers. ()
- 2- Windows is the most widely used operating system on computers. ()
- 3- The HTML programming language is easy to learn. ()
- 4- The Android operating system is an advanced operating system. ()
- 5- Apple iOS is an open-source operating system. ()
- 6- Android OS performs different basic functions than iOS. ()

2- Complete the following sentences using words between brackets:

(Code - Operating System - Apple iOS - Web Pages - Memory - Android)

- 1- The operating system controls computer functions, such as.....
- 2- Commands are given to the computer in the form of.....
- 3- Programmers use HTML to create.....
- 4-..... Cell phone operating system is closed source.
- 5- Most computers come preloaded with.....



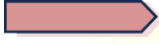
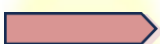
HTML



Hyper Text Markup Language (HTML) is the standard programming language used to create web pages.

The HTML tags

HTML consists of a series of basic elements to create a web page **such as:**

Titles  **Paragraphs**  **Links**  **Images**

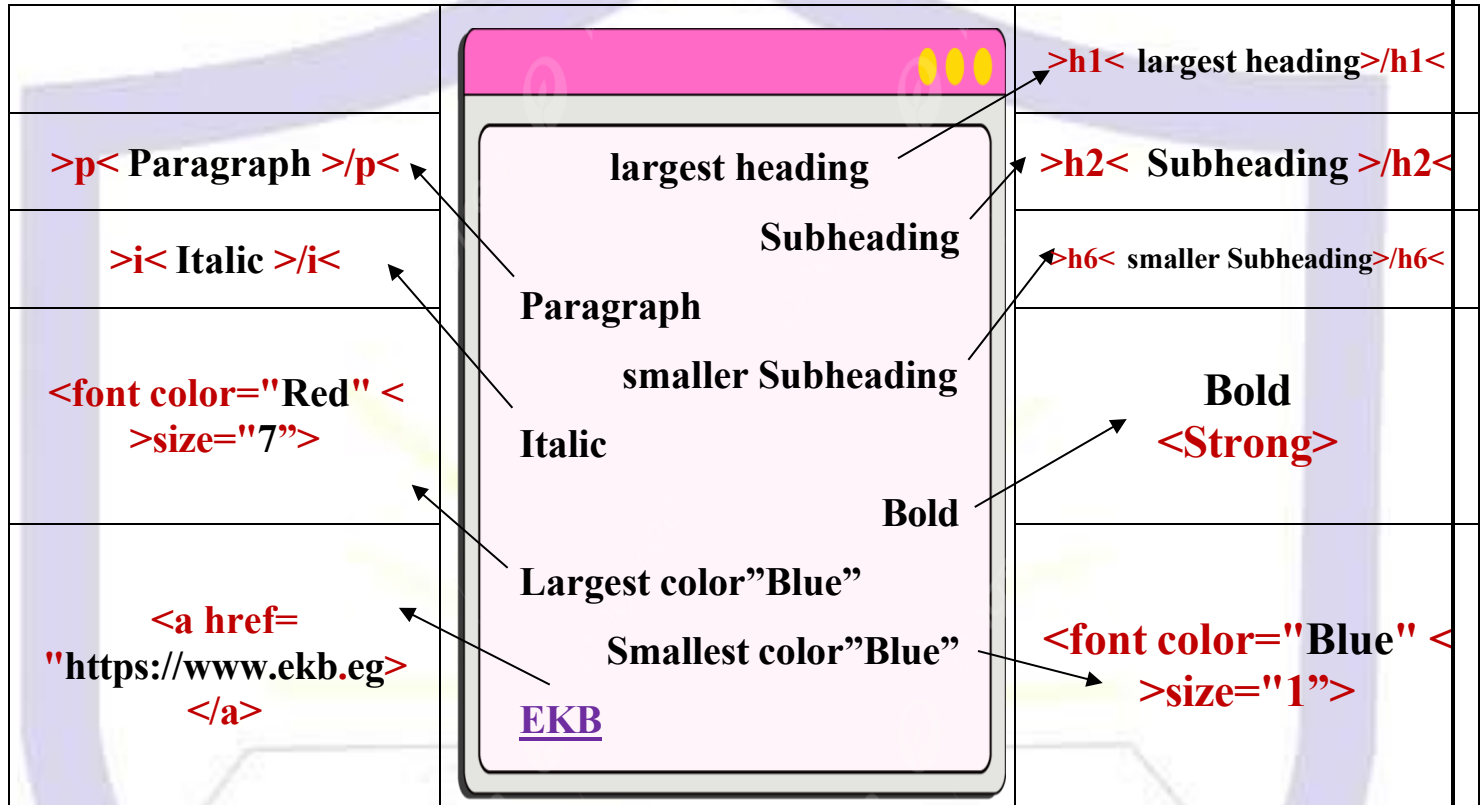
Web browsers read documents written in Hypertext Markup Language (HTML) and display the content according to the HTML tag.

The tags consist of three main parts:

- Opening Tag
- Content
- Closing Tag

```
1 <div class="box">
2   <pre>
3     <code>
4       <strong>Title:</strong>
5       <p>Code Rendered but not displayed</p>
6     </code>
7   </pre>
8 </div>
```

Hypertext Markup Language (HTML) components of a web page



Website designing features

- Hypertext markup language elements divide web pages to make them easier to look at, read, and organize.
- Hypertext markup language code also helps visually impaired people switch between parts of a webpage.

Images

attract attention and can trigger different emotions.

Paragraphs

Text broken up into paragraphs makes it easy for the reader to scan for the

Self-driving cars

Self-driving cars exist now, but it could be years before they are available to the public.

Self-driving cars and mobility

To aid mobility for blind and visually impaired individuals, cars can be designed with wide doors and flat floors as well as wheelchair accommodations.

A large heading

at the top of the page catches attention

Subheading

make the text easy to scan so a reader can find the information they want



Axis 1

Exercise

6



1-Put (✓) or (X):

1- HTML elements divide web pages to make them easier to look at, read, and categorize. ()

2- HTML tags consist of two main parts. ()

3- HTML code does not help the visually impaired distinguish between parts of a webpage. ()

4- Text divided into paragraphs makes it easier for readers to find the information they want. ()

5- A font that is difficult to read should be used in page design. ()

2- Choose the correct answer:

1- (HTML) is a language used to create.....

- (a) web pages (b) email (c) spreadsheets

2-..... attract attention and show different emotions in web pages.

- (a) small headings (b) images (c) italics

3- When creating your website, the content should be..... to make it easy for people to navigate.

- (a) difficult (b) important (c) unorganized (d) organized

4- At the top of the page, the attention-catched by

- (a) largest headline (b) images (c) paragraphs (d) video

5-..... Enables the reader to scan the text thoroughly to find the information they want.

- (a) headlines (b) subheadings (c) tags



Model exam(1)

1- Put (✓) or (X):

- 1- A modem is used to connect users to the Internet. ()
- 2- Augmented reality combines the real world with virtual reality. ()
- 3- Advanced technology helps people with disabilities to learn ()
- 4- Specialized search engines are considered a tool for digital research. ()
- 5- Smartphones are considered advanced operating systems. ()

2- Choose the correct answer:

- 1- is a language that describes the appearance of a web page
(a) I-stem (b) Android (c) HTML
- 2- is an Operating System For Apple computers only.
(a) Windows (b) Mac-Os (c) Android
- 3- The..... tag is used to write the largest address on a webpage
(a) <h1> (b) <P> (c) <a>
- 4- The program that controls the functions of the computer is known as.....
(a) Operating system (b) hacking (c) cyber intimidation
- 5- Connecting in wireless networks is done via.....
(a) ports (b) WI-Fi (c) Cables

3- Complete the following sentences using words between brackets:

(Image Capture - Screen Reader - Databases - User Experience - Binary)

1.is a good place to look for information
2.describes how people use and interact with the product
3.Automatically reads text uploaded
4. Mobile devices help to.....
5. Simple machine language is called System.....

Model exam (2)

1- Put (✓) or (X):

- 1- Paragraphs are only used at the top of the webpage. ()
- 2- Laptops are difficult to carry. ()
- 3- Smart gloves enable deaf people to translate their gestures into spoken output ()
- 4- Mac OS is one of the operating systems for cell phones. ()
- 5- The Egyptian Knowledge Bank is a public database, but it is considered unreliable . ()

2- Choose the correct answer:

1- is a collection of devices and objects linked together for common purposes

- (a) Networks (b) Computer (c) AI

2- The tag consists of..... parts.

- (a) two (b) three (c) four

3- The..... tag is used to add a paragraph.

- (a) (b) <P> (c) <a>

4- 4- When searching any database, use..... in the search

- (a) font color (b) numbers (c) keywords

5- Technology..... Enables people with disabilities to perform tasks they find difficult

- (a) old (b) inadequate (c) helpful

3- Match column A with the corresponding column B.

N	Column A		Column B
1	Artificial Intelligence	()	A- Combines the real world with virtual reality
2	Database	()	B-from specialized search engines
3	Virtual reality	()	C-a collection of information stored in a computer system
4	Google	()	D-A computer's ability to think, learn, and reason
5	Augmented reality	()	E-A virtual environment that allows users to explore and interact with their surroundings as if they were real.



Axes
2

The world of electronic games: fun and safety

1

Electronic games

» Electronic games which are played over the internet with friends, can be a lot of fun to play.

Advantages of online games

- They can help to promote critical and creative thinking
- Encourage teamwork.

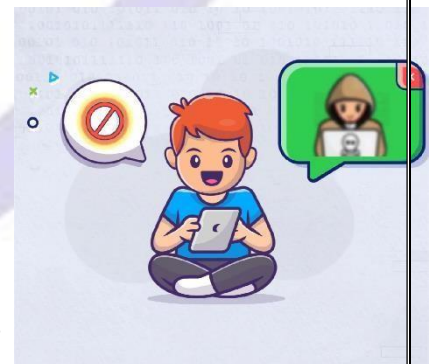


Disadvantages of electronic games

Online games can be fun to play .There are several disadvantages of electronic games

Security dangers

- Multiplayer games, in particular, can put your security at risk.
- If you share too much personal information, strangers may be able to access your Security dangers.
- Large gaming communities are filled with cyberbullies
- Hackers can cause your information to be leaked and viruses are easily spread.





Physical dangers

- Playing too often can affect your sleep cycle.
- It may keep you stimulated and prevent you from falling asleep.
- It can strain your eyes
- Cause back problems because you stay hunched over while playing, and cause headache



Psychological dangers

- Some players use inappropriate language or bully others
- which may generate aggressive feelings that affect studying, sleeping, and spending time with family and friends.



Playing video games responsibly

You can limit the dangers you are exposed to by:

- Only playing games and chatting with people you know and trust
- You should accept limits that your parents set to prevent unhealthy gaming habits.
- Make sure that gaming does not replace real-life activities like spending time with your family.
- When you do encounter dangers, don't be afraid to tell your parents, they want to help you.



Not using personal information in your username and passwords not using a webcam

 Respecting time slots can help you avoid negative online gaming habits.

Real-life activities mean spending time with family, meeting friends, doing homework, eating a healthy diet, exercising, or sleeping

The role of the family to minimize the impact of video games

Don't be afraid to turn to your parents whenever you encounter any risks,
they want to help you through:

- Family discussions
- Increased supervision when using the internet.



1- Put (✓) or (X):

- 1- Games help with critical thinking. ()
- 2- Video games encourage teamwork. ()
- 3- Playing too much video games affects your ability to sleep ()
- 4- You can share your name, email, and phone number online ()
- 5- Video games cannot strain one's relationship with one's family ()

2- Choose the correct answer:

1) Headaches and eye fatigue are.....risks

- (a) physical (b) emotional (c) psychological

2- Real-life activities should be practiced, such as:

- (a) exercise (b) online gaming (c) data sharing

3- Data sharing falls under the.....risks

- (a) ethical (b) psychological (c) security

4- Anxiety and depression are generated by risks

- (a) physical (b) psychological (c) ethical

5- One of the dangers of e-gaming is.....

- (a) teamwork (b) bullying (c) communication with friends

6- When playing online, you should only talk to people who are:

- (a) unknown and not trusted (b) unknown and trusted
(c) known and trusted



N.L.S

It is irresponsible to leave your devices at risk by not taking proper safety precautions.

Let's look at some more ways you can effectively protect your devices from theft.



Ways to protect digital devices from theft

1

PASSWORDS

Passwords with 8+ random letters, numbers, and symbols are the most secure.



2

Multi-Factor Authentication

MEFA requires two ways of identifying yourself, usually through passwords, pins, and security codes or personal questions.



3

Fingerprint AUTHENTICATION print

This type of authentication will scan your finger, (usually your thumb), as a way of confirming it's you.



4 FACIAL RECOGNITION

Its software will scan your face using your device's camera.

This way, only the user's face can be used to open their devices.



5 FULL ENCRYPTION

Encryption protects your information so only you can access it on your device, as long as you have set your password.



If you suspect your data has been compromised, you should:

-Report data theft.

-Notify a trusted adult or parent who can inform the authorities.

-Warn your friends and family members who communicate with you online about any hacking attempts.

Notice:



Notice:

1- Businesses especially need to protect their websites and their customers' information.

2- Egypt developed the Personal Data Protection Law to protect its citizens:

- Companies are required to use strict security foundations and standards to protect their data.
- Businesses must report immediately if they fall victim to data theft, and inform their customers directly as well.



1- Put (✓) or (X):

- 1- The password should preferably consist of numbers only. ()
- 2- Multi-factor authentication requires at least two applications for identification. ()
- 3- Devices are vulnerable if security measures are taken. ()
- 4- Fingerprint authentication can be used to confirm identity. ()
- 5- Businesses must protect their customers' data. ()
- 6- The facial recognition program will only perform an eye scan when the device is unlocked. ()

2- Complete the following sentences using words between brackets:

(Personal data protection - Fingerprint - Random symbols - data - Camera - Child)

- 1- Password that containsis more secure.
- 2- The facial recognition program performs a face scan using.....

3- If your on danger, be sure to tell your dad.

4- The law obliges companies to use strict standards to protect their data.

5- Your identity can be confirmed by.....



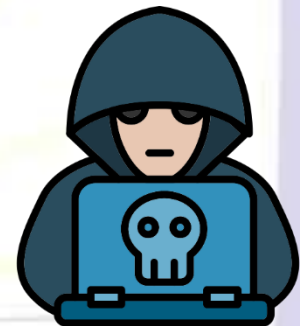
Axes
2

Be a cybersecurity hero

3

Cyber security

It is the protection of systems, networks, devices and data from cyber-attacks that may be exposed to them when using ICT applications.



Online safety precautions

-  Use safe web browsers and install anti-virus protection on your devices.
-  Use strong passwords and change them every three months.
-  Only allow friends and family to view your social media.
-  Verify websites that you frequent are real, safe, and age appropriate.
-  Verify information you find to make sure it is accurate.



Do not communicate with people you don't know or who make you feel uncomfortable.



Do not share personal information online.



Types of cyber security



Scareware

Is the most dangerous cyber threat, and you can fall victim to it no matter how careful you are?

You can be exposed to the threats of this malware when using unsecured Wi-Fi networks.

Examples

Scareware: This is a type of malware that takes the form of viral alarms, warning you to click on a link to get rid of the virus, but the link is actually the virus itself.



Smartphone malware.

Spooing

-Cyber criminals pretend to be a business, a known person, in order to provide your information, or click on unsafe links.

-If you click on unsafe links, you are in danger of

having your information compromised, or viruses installed on your devices.

-Examples of spoofing are a fake domain pretending to be a business, or an email account that has the name of a trusted service.

Examples

- Creating fake domains that mimic legitimate businesses.
- Creating an email account with the name of a trusted service or even an individual you know personally.



How to deal with cyber security threats

- Stay calm, but take swift action.
- Let your contacts know right away so they avoid opening anything suspicious. This way you can protect those you know from the same cyber threat.
- Use anti-virus software to do a sweep of your computer and remove any viruses.
- Confirm your security settings are up-to-date.
- Change your passwords - your device password, email, social media,
- Report the threat to a trusted adult.
- Creating a personal online safety plan can prepare you for potential future issues.



1- Put (✓) or (X):

- 1- it is Preferable to connect with people you don't know online. ()
- 2- Informing a trusted person when you are exposed to a cyber-threat is not important. ()
- 3- You can be exposed to malware, even if you are careful. ()
- 4- Making sure websites are safe and genuine is one way to stay safe online.()
- 5- Unsafe links can install viruses on your device. ()

2- Complete the following sentences using words between brackets:

(Antivirus - Cybersecurity - Insecure - Your Personal Information - Delete - Update)

- 1- Cybercriminals use.....to hack your online accounts.
- 2- Wi-Fi is considered..... Because it can expose you to cyber threats.
- 3- Take care to..... your security settings.
- 4- To delete any viruses, you should use programs
- 5- Protecting your data from being hacked is one of the important measures in.....



Copyright

Let's review what copyright (intellectual property rights) means.

- **" Intellectual property rights:"** is someone's invention, creation or idea, that is protected by law from being copied by someone else(individual , business).
- **Copyright** :includes written, visual, audio, video, and web material.
- **"Creative projects "** :refers to work that has been granted permission to share by the creator.
- Work in data libraries such as the Egyptian Knowledge Bank (EKB) is already approved for use.
- Egypt considers work in the public domain if the creator has been deceased for at least 50 years. No creator permission is required.

Intellectual property rights include



Written
content



Visuals



Audio



Videos



Materials from
the Internet

Stock exchange

- Is a big market where buyers and sellers deal in units of company stocks.
- These **stocks** are called shares.

Shares: These are small parts of a company's ownership.

For example, if you buy a share in a famous chocolate company, you become a small partner in it. If the company makes profits, you earn money along with it.



- The Egyptian stock exchange allows citizens to invest in any market sector, such as: food, clothing, or petroleum.
- Smart investors will research multiple companies before deciding which company to invest in.
- They analyze markets to see which companies' stocks go up or down.
- They will be checking to see the products or services offered, to see if it's a good fit for them personally.

A broker

This is a person who handles the buying and selling of shares

In traditional settings, an investor will have a broker

How to choose smart investors for your company

- They look for up-and-coming companies to invest in.
- They analyze markets to find out how competitive company stocks are.
- They study the products and services offered by companies and how well they align with their investment goals.

electronic financial trading

- E-trading has made the process of buying and selling shares more

through websites or applications.

- While investors can still use a broker, e-trading allows more independence and user-friendly options.
- **Robo-advisors:** are online applications that provide financial guidance and services.
- It provides access to many more people who otherwise might not be able to invest.
- It is easier for different companies to trade with one another faster, no matter where they are located.



1- Put (✓) or (X):

- 1- Works by an author who has been dead for 20 years do not need permission to be used. ()
- 2- Copyright includes written works only. ()
- 3- Works whose author has agreed to share them are referred to as a “creative project”. ()
- 4- Intellectual rights are a person's invention that is legally protected from being taken by someone else. ()
- 5- Facilitating the process of buying and selling company shares is one of the advantages of electronic financial trading. ()
- 6- There is no law in Egypt that protects the rights of innovators. ()

2- Choose the correct answer:

- 1- Works found at may be shared without permission.
- a) Creative project b) Digital books c) Internet d) Digital magazines

2- Company shares are called.....

- a) applications b) vehicles c) programs d) shares

3-..... are online applications that provide guidance and financial services to investors.

- a) Stocks b) Shares c) Advisory robots d) Companies

4- makes it easier for companies to execute trades quickly

- a) traditional b) outdated c) electronic financial d) unreliable

5- A stock exchange is defined as a marketplace where buying and selling is exchanged.....

- a) Websites b) Apps c) Consulting robots d) Company stocks

6- A marketplace where units of a company's stock are bought and sold is called.....

- a) Electronic market b) Stock exchange
c) Trading market d) Consulting robots



Axes
2

Cloud computing

5

Cloud computing

Is the on-demand storing, and accessing, of data and programs via remote servers on the internet.

It enables users to access files and data easily and conveniently from anywhere and at any time.

Examples of cloud computing



(**Drop box**)

From Drop box



(**Google Drive**)

From Google



(**one Drive**)

From Microsoft

» You can access OneDrive through **Microsoft 365**.

Remote servers enable users who are not connected to a local area network (LAN) to access and manage their data, files, and applications online.

Advantages of cloud

- 1- It frees up space on your devices by using cloud storage services.
- 2- It allows you to log in and access your files from multiple devices, as long as you have internet service.
- 3- It provides a backup for your files
- 4- It provides an extra layer of security by offering password protected data storage and data encryption.
- 5- It allows you to directly share files with authorized users

Disadvantages of cloud

- 1- It requires you to have an internet connection.
- 2- If you lose your internet connection, you won't be able to access files that are only stored in the cloud
- 3- Cloud storage is often a fee-based service.

Data encryption: The process of converting data into codes to prevent access by unauthorized parties.

Data Breach: Unauthorized parties accessing confidential data and information.



Storing files in the cloud

»» Cloud computing makes it possible to have a document management system to organize and easily find documents

Quickly, **by**:

- 1- Creating separate folders to save different data, such as photos, videos, audio files, and documents.
- 2- Separating school and personal information by creating a folder for each.
- 3- Clearly labeling documents and subfolders to make it easier to find what you're looking for.
- 4- Keep files in separate folders based on their types.

How to protect files on cloud storage platforms

- 1- Create strong passwords for your accounts and files on cloud storage platforms.
- 2- Use multi-factor authentication (MFA) for additional security.
- 3- Use large and reputable cloud storage services that offer multi-factor authentication.





1- Put (✓) or (X):

- 1- The storage capacity on your device can be augmented by using a cloud storage service. ()
- 2- Files stored in the cloud are accessed without an internet connection. ()
- 3- There is no risk in using cloud computing. ()
- 4- Cloud computing provides backup copies of files. ()
- 5- Cloud computing provides an additional layer of security. ()
- 6- Files stored on the cloud can be accessed from only one device. ()
- 7- It is preferable to separate school and personal information on the cloud. ()

2- Complete the following sentences using words between brackets:

(Google Drive - Remote Servers - Internet Connection - Cloud Computing - Document Management)

- 1-..... enables users who are not connected to the local network to access their files over the Internet.
- 2- Cloud computing follows thesystem to organize and find your documents easily and quickly.
- 3- is an example of cloud computing.
- 4- Accessing files stored on the cloud requires.....

5- A technology that allows data to be stored and accessed through remote servers over the Internet is.....



Axes
2

CLOUD COMPUTING APPLICATIONS

6

Cloud Computing

Cloud computing is storing, and accessing, data or programs through the internet.

The most popular software that deal with cloud computing applications:

- Microsoft 365
- Google Workspace



Microsoft 365 productivity Apps

6

Out Look



5

One Drive



**Microsoft
Excel**



3

**Microsoft
PowerPoint**



2

**Microsoft
Word**



**Microsoft
Teams**



- 1- **Word** can be used to jot down your ideas, a story, or a school assignment.
- 2- **PowerPoint** helps to create a presentation.
- 3- **OneDrive** helps to share your work with your colleagues.
- 4- **Outlook** is an email application.

OneDrive

OneDrive can help you share work with your friends.

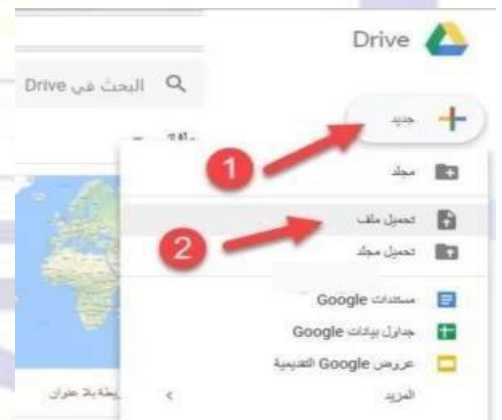
- OneDrive allows you to store, share, and access files from any device.
- It's very user- friendly.
- You upload your files and then click on the Share option to share.
- You can decide what level of access each person you share with gets (viewing purposes only, or editing).



How to Upload Files or Folders?

» You can upload files to OneDrive with the following steps:

- 1- Open the Google Chrome browser.
- 2- Sign in to Microsoft Office.
- 3- Select OneDrive.
- 4- Choose Upload, then Upload Files or Folders.
- 5- Select the files or folders to be uploaded.
- 6- Select Upload, then Confirm Upload.



How to Save Files or Folders?

- 1- Choose a file, then save a copy to OneDrive either for company or personal



3- Choose Save



6



- 2- Complete the following sentences using words between brackets:**

- 1- Theapp is used to create a school report.
- 2- To make a file share, you must first..... Files on OneDrive
- 3- is a platform for email.
- 4- helps to share your work with your colleagues.

5- You can see the files after saving them to OneDrive when you open the app



Model exam(1)

1- Put (✓) or (X):

- 1- Playing too much puts you at risk. ()
- 2- A strong password consists of only four characters. ()
- 3- Do not verify the information you find online. ()
- 4- Copyrights include written, visual and audio works. ()
- 5- Cloud computing allows access to data at any time ()

2- Choose the correct answer:

1- Protecting systems, networks, devices and data from cyber-attacks are called

- (a) Hacking (b) Penetration testing (c) Cybersecurity

2- One of the disadvantages of massively multiplayer online games is.....

- (a) knowing friends (b) security risks (c) e-learning

3-..... Requires at least two ways to identify yourself

- (a) multi-factor authentication (b) fingerprint (c) facial recognition

4-..... refers to works whose creators have given permission to share them.

- (a) Creative Commons (b) Piracy (c) Phishing

5-allows the user to access files anywhere easily

- (a) file downloading (b) cloud computing (c) artificial intelligence

3- Complete the following sentences using words between brackets:

(Artificial Intelligence - Wi-Fi - Encryption - Word - One Drive)

1. Allows you to store and share files
2. Use..... to unlock your cell phone using your fingerprint
3. Your cell phone can be connected to the network via.....
4. The program allows you to allows you to write down your notes
5. protects your information so no one can access it

Model exam (2)

1- Put (✓) or (X):

- 1- Video games encourage teamwork and do not create any risks. ()
- 2- Multi-factor authentication is one of the appropriate security methods to protect your digital devices. ()
- 3- Scareware is an example of malware. ()
- 4- Electronic financial trading facilitates the buying and selling of stocks between investors face-to-face ()
- 5- The Outlook application is used to send email ()

2- Choose the correct answer:

2- Using Wi-Fi..... networks Exposes you to malware threats

(a) insecure (b) local (c) closed

3- Multi-factor authentication requires..... at least

(a) three ways (b) four ways (c) two ways

4- Anxiety and depression are generated by the..... risks for electronic games.

(a) security (b) psychological (c) physical

5- The..... program uses your camera to unlock your device

(a) passwords (b) encryption (c) facial recognition

3- Match column A with the corresponding column B.

N	Column A	Column B	
1	Stock Exchange	()	A- An assistive technology tool for people with disabilities
2	Intellectual rights	()	B-protects your information so that no one can access it
3	Smart gloves	()	C-An app that provides investors with financial guidance and services
4	encryption	()	D-A marketplace where buyers and sellers exchange units of a company's stock
5	Counseling robots	()	E-Someone's invention, protected by law from being taken by someone else.

Model exam on Axis 1,2

1- Put (✓) or (X):

- 1- There are two types of networks to create a connection between computers. ()
- 2- User experience is related to the development of products and services. ()
- 3- HTML tags consist of four main parts ()
- 4- There are no advantages to playing online games ()
- 5- Full encryption works to protect your information ()

2- Choose the correct answer:

- 1-is at the top of the webpage and draws attention
(b) smallest title (b) largest title (c) subtitle
- 2- The beginning tag in HTML takes the form.....
(C) <> (b) </> (c) Content
- 3- Does not include copyrights
(a) videos (b) written materials (c) public domain
- 4- Creating backups that protect you from digital hacking is called.....
(a) Cybersecurity (b) Phishing (c) Impersonation

1- Complete the following sentences using words between brackets:

(Google Drive - Modem - OneDrive - Egyptian Knowledge Bank - Virtual Reality - iOS)

- 1-.....is considered an advanced technology
- 2-..... is a component of a computer network
- 3-is the operating system used on Apple mobile devices
- 4- One of the applications of Microsoft 365.....
- 5- is considered one of the platforms that allow the use of the works and books in it

